SentinelAI Enterprise

60-Day NVIDIA-Focused Engineering Roadmap

Vision

Build the most complete open-source AI Risk Monitoring and AI Infrastructure Observability Platform for modern enterprise LLM systems.

Goal:

Enable organizations to monitor, secure, benchmark, audit, and govern AI applications running on large language models.

Think:

LangSmith + Datadog + NVIDIA NIM Guardrails + Grafana

combined into a single platform.

Why This Project

Most AI projects today focus on:

- Chatbots
- RAG systems
- AI agents
- Productivity tools

Very few focus on:

- AI Security
- AI Infrastructure
- Model Observability
- AI Governance
- GPU Performance Analytics

These are exactly the categories attracting significant enterprise investment.

This project demonstrates:

- Distributed systems thinking
- AI engineering

- Security engineering
- ML infrastructure
- Full-stack architecture
- Production-grade software development

Product Name

SentinelAI Enterprise

Tagline:

"Observability, Security and Governance for Enterprise AI."

Core Product Modules

Module 1: AI Security Layer

Purpose:

Protect AI applications from malicious inputs and unsafe outputs.

Features:

Prompt Injection Detection

Detect:

- Ignore previous instructions
- System prompt extraction
- Hidden instructions
- Context poisoning

Output:

- Risk score
- Attack category
- Explanation

Jailbreak Detection

Detect:

- DAN attacks
- Roleplay attacks
- Character attacks
- Multi-turn manipulation

Output:

- Severity score
- Confidence score
- Recommended action

PII Leakage Detection

Detect:

- Email addresses
- Phone numbers
- Credit card numbers
- API keys
- Secrets

Output:

- Redacted response
- Leakage report

Toxicity Detection

Detect:

- Hate speech
- Harassment
- Toxic content
- Unsafe generations

Output:

- Moderation score
- Policy violation report

Module 2: AI Observability Layer

Purpose:

Provide complete visibility into AI application behavior.

Features:

Request Monitoring

Track:

- Request volume
- User activity
- Model usage

Response Monitoring

Track:

- Response time
- Token usage
- Completion quality

Latency Analytics

Track:

- Average latency
- P95 latency
- P99 latency

Cost Analytics

Track:

- OpenAI cost
- Anthropic cost
- Gemini cost
- Self-hosted model cost

Module 3: Hallucination Detection

Purpose:

Measure factual reliability.

Methods:

Retrieval Verification

Compare response against:

- Knowledge base
- Retrieved documents

Generate:

Hallucination score

Confidence Evaluation

Calculate:

- Confidence score
- Risk score

Generate:

Trustworthiness report

Module 4: Multi-Model Evaluation Lab

Purpose:

Benchmark LLMs under identical workloads.

Models:

- Llama
- Qwen
- Gemma
- DeepSeek
- Mistral

Evaluation Areas:

- Latency
- Throughput
- Accuracy
- Cost
- Hallucination Rate

Outputs:

- Leaderboards
- Performance reports
- Evaluation dashboards

Module 5: GPU Observability

Purpose:

Provide NVIDIA-focused infrastructure monitoring.

Metrics:

GPU Usage

Track:

- Utilization
- Temperature
- Power draw

Memory Analytics

Track:

- VRAM consumption
- Memory spikes
- OOM risks

Inference Analytics

Track:

- Tokens/sec
- Throughput
- Concurrent requests

Module 6: Enterprise Workspaces

Purpose:

Enable team collaboration.

Features:

Teams

- Create organizations
- Create workspaces
- Invite members

Roles

- Owner
- Admin
- Security Analyst
- Developer
- Viewer

Audit Logs

Track:

- User actions
- Model changes
- Security incidents

Module 7: Incident Management

Purpose:

Enterprise-grade AI risk operations.

Features:

Incident Creation

Automatically create incidents when:

- Prompt injection detected
- Hallucination exceeds threshold
- Data leakage occurs

Incident Dashboard

Track:

- Open incidents
- Severity
- Status
- Resolution time

Technical Architecture

Frontend:

Next.js TypeScript TailwindCSS Shadcn UI Framer Motion

Backend:

FastAPI

Services:

Auth Service Analysis Service Security Service Benchmark Service Workspace Service Incident Service

Database:

PostgreSQL

Cache:

Redis

Queue:

Celery

Storage:

MinIO

Monitoring:

Prometheus Grafana

Deployment:

Docker Kubernetes

Cloud:

AWS / GCP

GitHub Repository Structure

sentinel-ai-enterprise/

frontend/

backend/

services/

security-engine/

hallucination-engine/

benchmark-engine/

gpu-monitor/

workspace-service/

incident-service/

docs/

architecture/

benchmarks/

research/

Deliverables

By Day 60

1. Production-grade SaaS platform
2. Open-source GitHub repository
3. Technical architecture diagrams
4. Engineering blog series
5. Benchmark reports
6. Security research reports
7. Public product demo
8. Documentation website
9. Docker deployment
10. Kubernetes deployment

Weekly Timeline

Week 1

Architecture Design Database Design System Design

Week 2

Security Engine

Week 3

Observability Engine

Week 4

Hallucination Engine

Week 5

GPU Monitoring

Week 6

Enterprise Workspace

Week 7

Incident Management

Week 8

Polishing Documentation Deployment Launch

NVIDIA Attention Strategy

After launch:

Step 1

Publish:

"Building SentinelAI Enterprise: Open Source AI Security & Observability Platform"

Medium Dev.to Hashnode

Step 2

Create:

Architecture diagrams

System design breakdowns

Engineering blogs

Step 3

Contribute to:

NVIDIA NeMo

TensorRT

Triton Inference Server

CUDA Samples

Step 4

Create benchmarking reports comparing:

Llama Gemma Qwen DeepSeek

on NVIDIA GPUs.

Final Goal

Become known as:

"The engineer building infrastructure for AI systems"

instead of

"The student building AI applications."